

Camouflage Unlearning: Fast and Surgical Machine Unlearning for Biometric Systems

Abstract—Machine unlearning for privacy-critical biometric systems such as fingerprint, iris, and face recognition remains underexplored, despite the growing need for efficient solutions under General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) mandates that require data erasure requests to be supported throughout a model’s deployment lifecycle. Existing methods predominantly target classification and generative tasks, and rely on many-to-one feature collapse that forces all forget classes toward a single distant point in feature space. This increases convergence time and computational cost, making them impractical for resource-constrained edge deployments where biometric models face frequent sequential removal requests with minimal replay data. Additionally, existing methods have been benchmarked almost exclusively on ResNet or ViT backbones for CIFAR and ImageNet, leaving their effectiveness on biometric backbones largely untested. In this work, we propose Camouflage Unlearning (CU), a cycle-robust framework that maps forget classes toward nearby retain-class boundaries rather than collapsing to a single point, thereby reducing migration distance. CU achieves 3–4 $\times$ faster convergence through retain-dominance constraints and localized regularization that performs surgical updates in specific transformer blocks, while maintaining competitive retain accuracy. Evaluation across four domains, classification, iris, fingerprint, and face recognition demonstrates competitive performance and privacy behavior (MIA $\approx$ 0.5), supporting CU’s effectiveness for biometric authentication systems requiring regulatory compliance.

Index Terms—Biometric recognition, data privacy, face recognition, GDPR, machine unlearning, vision transformers

I. INTRODUCTION

MACHINE unlearning (MU) has emerged as a critical research area, addressing the imperative to prevent unauthorized data exploitation, eliminate harmful model behaviors, and safeguard user privacy [1]. A natural consideration follows: what drives the necessity for unlearning? The proliferation of AI misuse ranging from phishing generation and model inversion attacks to perpetuating misinformation has prompted regulatory frameworks [2], [3], [4]. Frameworks like the General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) [5], [6] mandate data erasure capabilities. These regulations require model service providers to erase user data from learned representations upon request. This establishes both legal and ethical obligations for responsible AI deployment. Crucially, users possess the right to request data removal at any time. This transforms unlearning from a one time operation into a continuous process throughout a model’s deployment lifecycle. Per-request efficiency, therefore, becomes a first-class concern.

Existing machine unlearning methods predominantly target image classification [7] and generative tasks [8], with limited exploration of privacy-critical biometric systems. While face

recognition [9], [10], [11] has received modest attention, fingerprint and iris recognition remains substantially less studied. This gap persists despite their sensitive nature and widespread deployment in authentication systems. Contemporary approaches rely primarily on KL divergence maximization [12] or gradient ascent [13]. These methods implement many-to-one mappings that force forgotten class embeddings toward a single distant point in feature space. Alternative strategies include saliency-based selection [14], geometric boundary manipulation [15], perturbation-based forgetting [16], and meta-learning optimization [17], [18]. A key observation across all these methods is that every forget sample is driven toward a single collapse point in the decision space. Even when a sample already lies far from the retain manifold, it must still traverse the full distance to this distant target. This increases convergence time and GPU cost, as observed in Table I. This many-to-one collapse behavior is visualized and ablated in detail in Section III-B. Concretely, GS-LoRA [13] (group-sparse LoRA, slow ascent), SCRUB [12] (bidirectional logits, unstable KL), SalUn [14] (weight saliency, reversible labels), Boundary Unlearning [15] (shadow-node pruning, slow), Forget Vectors [16] (noise tuning, invertible), MCU [18] (Bézier curves, dual optimization), LTU [17] (meta-learning, computationally expensive), DELETE [19] and LoTUS [20] (frozen teacher, double memory), and MUNBa [21] (Nash bargaining, per-step overhead) each trade convergence speed, memory, or reversibility, and few explicitly target biometric edge deployment or sequential forget cycles.

Biometric recognition differs from general classification in ways that directly affect unlearning feasibility. Biometric systems deploy on resource-constrained edge devices. Removal requests arrive sequentially and frequently over the deployment lifecycle. This transforms unlearning into a recurring operational cost rather than a one-time procedure. Under this workload, existing many-to-one methods become prohibitive as per request compute scales poorly with request frequency. Compounding this, existing unlearning methods have been designed and benchmarked almost exclusively on ResNet [22] or ViT backbones [23] for CIFAR [24] and ImageNet [25]. Their transferability to biometric backbones such as Face Transformer [26] (sliding-patch ViT face recognition), LAM [27] (ResNet-50, lightweight attention), JIPNet [28] (CNN-Transformer joint pose-identity), SSLDMNet-Iris [29] (CNN-GRU, Fisher discriminant), IrisFormer [30] (rotation-invariant ViT matching), PFSimNet [31] (partial fingerprint similarity network), and MEM-FR [32] (optical, privacy-preserving recognition) remains untested, despite these architectures being the standard in deployed authentication systems. What biometric unlearning requires instead is a cycle-robust method validated directly on biometric backbones.

To address these challenges, we propose Camouflage Unlearning (CU), a novel framework that achieves 3–4 $\times$ faster convergence while maintaining robust performance. Unlike existing many-to-one approaches, CU pushes forget-class embeddings toward the nearest retained class boundaries, effectively reducing migration distances. To perform surgical modifications and minimize collateral damage to retain classes, CU employs localized regularization concentrated in specific blocks of the neural network for additional details, please refer to Section II.

Our main contributions are summarized as follows:

- 1) To the best of our knowledge, this is the among the earliest comprehensive studies of machine unlearning for privacy-critical biometric systems, including fingerprint, iris, and face recognition.
- 2) We introduce Camouflage Unlearning, which achieves 3–4 $\times$ faster convergence compared to SoTA unlearning methods while maintaining competitive retention accuracy and privacy guarantees.
- 3) We conduct extensive empirical evaluation across multi-ple biometric and classification domains, demonstrating that CU concentrates updates in specific network blocks and remains effective under data-scarce replay conditions.

II. PROBLEM SETTING AND METHODOLOGY

A. Problem Setting

Let $\mathcal{M}$ be a model pre-trained on $D = D_r^{full} \cup D_f$ ($D_r^{full} \cap D_f = \emptyset$), representing a mapping $f_{\mathcal{M}} : \mathcal{X}_D \rightarrow \mathcal{Y}_D$. Storing D_r^{full} is prohibited under GDPR/CCPA, and full retraining is costly, so we retain only a small replay buffer $D_r \subset D_r^{full}$, with $|D_r|$ limited to at most 10% of $|D_r^{full}|$ ¹. Before unlearning, $\mathcal{M}$ correctly maps both D_f and D_r . Applying an unlearning algorithm $\mathcal{F}$ (Section II-B) yields $\mathcal{M}' = \mathcal{F}(\mathcal{M}, D_f, D_r)$, under which the forget mapping should no longer hold while the retain mapping persists:

$$f_{\mathcal{M}} : \mathcal{X}_{D_f} \rightarrow \mathcal{Y}_{D_f}; \mathcal{X}_{D_r} \rightarrow \mathcal{Y}_{D_r} \implies f_{\mathcal{M}'} : \mathcal{X}_{D_f} \not\rightarrow \mathcal{Y}_{D_f}; \mathcal{X}_{D_r} \rightarrow \mathcal{Y}_{D_r} \quad (1)$$

where $\not\rightarrow$ denotes a forgotten mapping and $\rightarrow$ a retained one. We define successful unlearning as

- Forgetting: $Acc(\mathcal{M}', D_f) \leq \frac{1}{C}$,
- Retention: $Acc(\mathcal{M}', D_r) \approx Acc(Oracle, D_r)$

where C is the number of classes and Oracle denotes a model trained directly on target classes only.

B. Camouflage Unlearning

CU relocates forget-class embeddings to retain-class boundaries via direct centroid mapping. Unlike existing many-to-one approaches that collapse all forget classes to a single distant point, CU implements many-to-many distribution (Figure 1), reducing migration distance and accelerating convergence.

We compute class centroids by averaging embeddings, $c_k = \frac{1}{|D_k|} \sum_{x \in D_k} emb(x)$, yielding retain centroids $\{c_{r_1}, \dots, c_{r_K}\}$ and forget centroids $\{c_{f_1}, \dots, c_{f_M}\}$, where K and M are the

numbers of retain and forget classes. Each forget centroid c_{f_i} is assigned its nearest retain centroid as target:

$$\mathbf{t}_i = c_{r_{j^*}}, \quad j^* = \arg \min_j \|c_{f_i} - c_{r_j}\|_2 \quad (2)$$

directly mapping forget embeddings into retain decision boundaries and minimizing migration distance. We pull forget embeddings toward their assigned targets:

$$\mathcal{L}_f = \frac{1}{|D_f|} \sum_{(\mathbf{x}, y) \in D_f} \|emb(\mathbf{x}) - \mathbf{t}_y\|^2 \quad (3)$$

which collapses forget classes into retain boundaries but risks corrupting retain decision regions and prolonging convergence. We therefore enforce that retain logits dominate forget logits by margin $\gamma = 2.0$:

$$\mathcal{L}_{dom} = ReLU \left(\max_{i \in F} z_i + \gamma - \max_{j \in R} z_j \right) \quad (4)$$

where z_i are logits and F, R denote forget and retain class indices; this accelerates forget-class suppression while preserving retain accuracy. Retain-class knowledge is preserved through classification and embedding stability:

$$\mathcal{L}_{ret} = \lambda_{ce} \cdot CE(\mathbf{z}_r, y_r) + \lambda_{emb} \cdot \frac{1}{|D_r|} \sum_{(\mathbf{x}, y) \in D_r} \|emb(\mathbf{x}) - c_y\|^2 \quad (5)$$

where $\mathbf{z}_r$ are retain logits, y_r are labels, and c_y are original retain centroids. Since comprehensive parameter updates risk instability under limited replay data, inspired by group sparse selection [13], we apply localized regularization that selectively updates fewer transformer blocks:

$$\mathcal{L}_{local} = \lambda_{local} \sum_{g=1}^G \|\theta_g\|_2 \quad (6)$$

where θ_g represents parameters in block g and G is the total number of groups. Each θ_g is initialized at zero and tracks the *change* accumulated in block g relative to the pretrained weights, so $\mathcal{L}_{local}$ penalizes the magnitude of updates rather than absolute parameter magnitude. This group-Lasso-inspired approach concentrates updates in specific blocks while leaving others unchanged, preserving retained knowledge under constrained replay. The total loss integrates all components:

$$\mathcal{L}_{total} = \mathcal{L}_f + \lambda_{dom} \mathcal{L}_{dom} + \mathcal{L}_{ret} + \mathcal{L}_{local} \quad (7)$$

C. Algorithmic Preview

Algorithm 1 summarizes the complete procedure: computing class centroids, assigning each forget class to its nearest retain centroid as a target, and iteratively updating the model using the combined forget, dominance, and retain losses.

III. PERFORMANCE EVALUATION

A. Experimental Setup

Datasets and Backbones: To comprehensively evaluate the proposed CU, we examine its performance across four distinct tasks. We begin with image classification on CIFAR-100 [24], employing the DeiT-Tiny model [33]. Extending

¹From this point, D_r denotes the replay buffer unless specified.

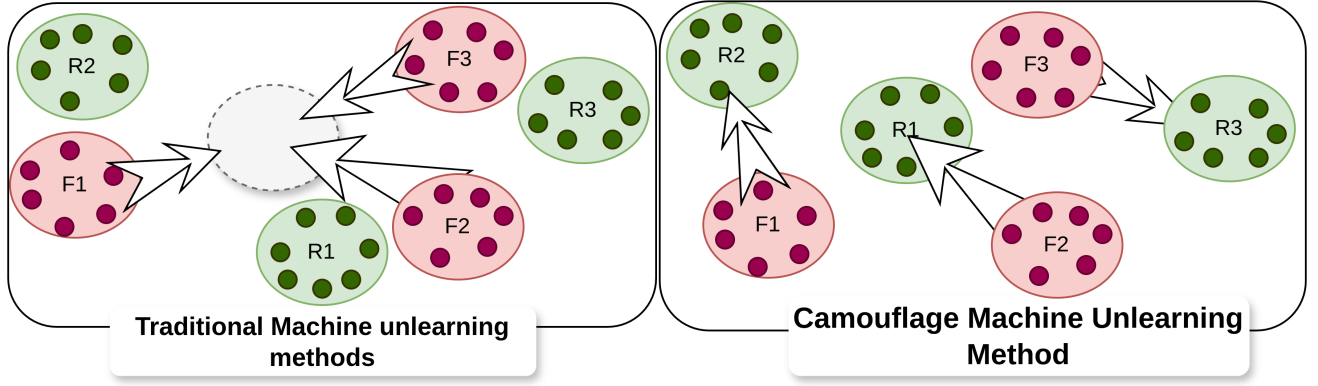


Fig. 1. Overview of Camouflage Unlearning. Traditional methods collapse all forget classes to a single distant point, requiring long migration paths. CU instead hides each forget class within its nearest retain class boundary, reducing migration distance and accelerating convergence.

Algorithm 1 Camouflage Unlearning

Require: Pre-trained model $\mathcal{M}$, forget data D_f , retain data D_r , margin γ , hyperparameters $\lambda_{\text{dom}}, \lambda_{\text{ce}}, \lambda_{\text{emb}}, \lambda_{\text{local}}$, epochs E

Ensure: Unlearned model $\mathcal{M}'$

```

1: for each class  $k$  in  $D_r \cup D_f$  do
2:    $c_k \leftarrow \frac{1}{|D_k|} \sum_{x \in D_k} \text{emb}(x)$ 
3: end for
4: for each forget class  $i$  with centroid  $c_{f_i}$  do
5:    $j^* \leftarrow \arg \min_j \|c_{f_i} - c_{r_j}\|_2$ ;  $t_i \leftarrow c_{r_{j^*}}$   $\triangleright$  Eq. 2
6: end for
7: for epoch = 1 to  $E$  do
8:   for each mini-batch from  $D_f, D_r$  do
9:      $\mathcal{L}_f \leftarrow \frac{1}{|D_f|} \sum_{(x,y) \in D_f} \|\text{emb}(x) - t_y\|^2$   $\triangleright$  Eq. 3
10:     $\mathcal{L}_{\text{dom}} \leftarrow \text{ReLU}(\max_{i \in F} z_i + \gamma - \max_{j \in R} z_j)$   $\triangleright$ 
Eq. 4
11:     $\mathcal{L}_{\text{ret}} \leftarrow \lambda_{\text{ce}} \cdot \text{CE}(\mathbf{z}_r, \mathbf{y}_r)$ 
       $+ \lambda_{\text{emb}} \cdot \frac{1}{|D_r|} \sum_{(x,y) \in D_r} \|\text{emb}(x) - c_y\|^2$   $\triangleright$ 
Eq. 5
12:     $\mathcal{L}_{\text{local}} \leftarrow \lambda_{\text{local}} \sum_{g=1}^G \|\theta_g\|_2$   $\triangleright$  Eq. 6
13:     $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_f + \lambda_{\text{dom}} \mathcal{L}_{\text{dom}} + \mathcal{L}_{\text{ret}} + \mathcal{L}_{\text{local}}$   $\triangleright$  Eq. 7
14:    Update  $\mathcal{M}$  via  $\nabla_{\theta} \mathcal{L}_{\text{total}}$ 
15:   end for
16: end for
17: return  $\mathcal{M}'$ 

```

our evaluation to biometric authentication, we examine iris recognition using CASIA-Iris [34] with the lightweight LAM architecture [27]. For fingerprint verification, we leverage the Sokoto Coventry dataset [35] paired with JIPNet [28], a feature learning approach that eliminates hand-crafted descriptors. Our evaluation concludes with face recognition on CASIA-Face100 [36] where we deploy the Face Transformer [26] as our recognition backbone.

Baselines: To establish a comprehensive benchmark, we compare against established and recent machine unlearning methods. From foundational approaches, we implement SCRUB [12], a student-teacher framework with bidirectional

logit manipulation; SalUn [14], which employs gradient-based weight saliency; and GS-LoRA [13], utilizing group sparse regularization with LoRA adapters. We further evaluate emerging methods: LTU [17], applying meta-learning with gradient harmonization; MCU [18], leveraging Bézier curve optimization in parameter space; and Forget Vectors [16], which perturbs inputs to suppress forget class predictions. All methods are evaluated against an Oracle baseline, defined as a model trained exclusively on D_r^{full} , representing the best achievable performance on retain classes. This provides a rigorous evaluation framework for our proposed CU method.

Evaluation Metrics: We evaluate forget accuracy Acc_f , retain accuracy Acc_r , membership inference attack success (MIA), KL divergence, and runtime efficiency (RTE), measured in seconds. Following Shibata *et al.* [37], we use H-mean as $H\text{-Mean} = \frac{2 \cdot \text{Acc}_r \cdot (1 - \text{Acc}_f)}{\text{Acc}_r + (1 - \text{Acc}_f)}$. Successful unlearning requires $\text{Acc}_f \approx 0$, $\text{MIA} \approx 0.5$ indicates near-random membership inference, low KL divergence and RTE, with Acc_r near oracle performance. **MIA follows the label-only attack of [38], combining gap, boundary-distance, and augmentation attacks, and reports attack accuracy on member (forget-train) versus non-member data, where 0.5 indicates the attacker cannot infer membership. KL divergence is reported as $\text{KL}(\mathcal{M}_{\text{unlearned}} \parallel \mathcal{M}_{\text{oracle}})$ over the evaluation set, measuring proximity of the unlearned model to the oracle. Because class-attribution accuracy alone does not capture the open-set, embedding-distance-thresholding nature of real-world biometric verification, we additionally report False Accept Rate (FAR) at thresholds 0.3, 0.4, and 0.5, and Equal Error Rate (EER), computed by matching forget-class probes against the retain gallery under cosine similarity; low FAR/EER close to the Oracle indicates that removed identities are not confused with any retained identity beyond chance level (Section III-D).**

Implementation Details: All experiments are conducted on an NVIDIA RTX A4000 GPU using PyTorch [39] and Python 3.11. Unless otherwise specified, we set $\lambda_{\text{dom}} = 0.1$, $\lambda_{\text{ce}} = 0.4$, $\lambda_{\text{emb}} = 1.0$, and $\lambda_{\text{local}} = 0.01$, with margin $\gamma = 2.0$. All baselines and CU are fine-tuned for 10 epochs using SGD [40]

with a learning rate of 10^{-4} , momentum of 0.9, and batch size of 64.

B. Main Results

Table I presents results across four tasks: classification, iris recognition, fingerprint recognition, and face recognition, where **green** highlights best performance and **blue** indicates second-best.

Unlearning and Retention: All methods achieve near perfect forgetting $Acc_f < 1\%$, approaching random assignment. Proposed CU consistently ranks first or second in Acc_r across all tasks, with LTU as the strongest competitor on classification, iris, and fingerprint tasks, while MCU excels on iris recognition. For $H\text{-Mean}$, proposed CU alternates between first and second positions alongside competitive showings from LTU, FV, MCU, and GS-LoRA.

Privacy and Efficiency: GS-LoRA achieves lower KL divergence than CU, which is expected since CU deliberately modifies logit distributions to suppress forget classes and is therefore less competitive on this metric. However, MIA scores near 0.5 confirm CU provides comparable privacy protection. Most critically, proposed CU achieves 3 to 4 $\times$ speedup versus all baselines, validating our design objective: fast, privacy-preserving unlearning without catastrophic forgetting.

Geometric Analysis of Unlearning Behavior: To understand geometric behavior across unlearning methods, we visualize embedding space transformations using t-SNE. Figure 2 compares baseline methods against proposed CU. Existing approaches GS-LoRA [13], SCRUB [12], and Boundary Unlearning [15] collapse forget classes into a single distant point, requiring long distance traversal in feature space. CU distributes forget samples across nearest retain class boundaries, achieving faster convergence through shorter migration paths, explaining the RTE gains in Table I.

C. Ablation Studies

Replay Buffer Size: We investigate the minimal replay buffer size $|D_r|$ for effective unlearning on CIFAR-100. Table II shows that a 0.1 ratio optimally balances retention accuracy and efficiency.

Retain Dominance Constraint: Without this constraint, we found that the forget accuracy exhibits a prolonged tail, delaying convergence. Table III demonstrates that applying the constraint reduces Acc_f from 9.97% to 0.39%, achieving faster unlearning with minimal retention loss.

Localized Parameter Regularization: This regularization selectively modifies fewer blocks, enabling surgical edits that preserve model capacity. On CIFAR-100 classification, Figure 3 shows concentrated updates in specific blocks, while Table IV demonstrates reducing total ℓ_2 norm from 39.05 to 10.09 with minimal retention loss.

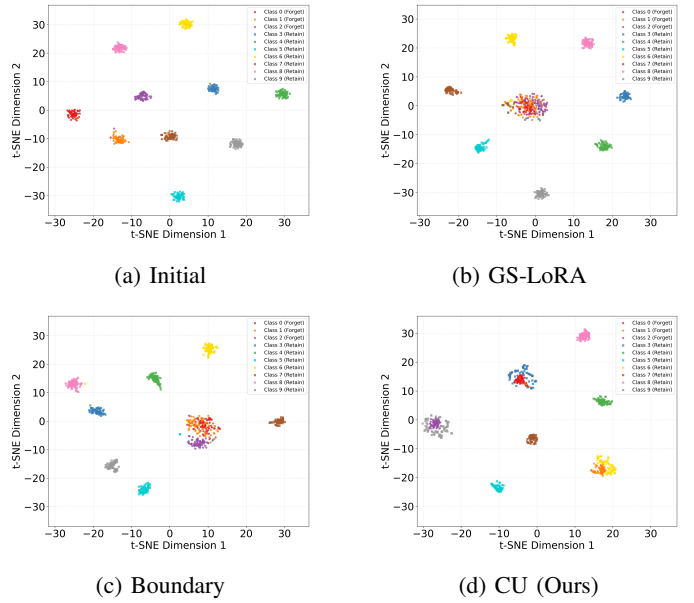


Fig. 2. t-SNE visualization. Baselines collapse to single point while proposed CU method distributes to nearest boundaries thus achieving faster convergence.

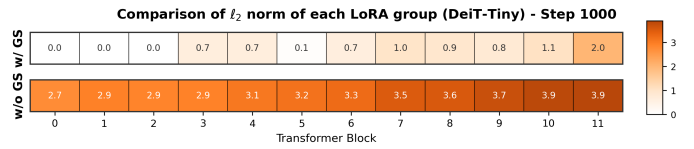


Fig. 3. ℓ_2 norm per block. Localized regularization concentrates updates in specific blocks.

D. Attack-Based Verification

Open-Set Verification Metrics: Table V reports FAR/EER of forget-class probes matched against the retain gallery on CASIA-Iris. CU tracks the Oracle closely (FAR@0.5 3.83% vs. 3.10%; EER 24.99% vs. 21.69%), and both rise similarly over *Before* unlearning (1.64%), indicating that the small increase in open-set false accepts stems from removing the identities rather than from CU specifically. Whether the residual accepts concentrate on individual retain identities is a separate question, addressed per-identity below.

Per-Identity Confound Analysis: A natural concern is whether mapping a forget identity to its nearest retain centroid creates a targeted, per-identity vulnerability rather than a general accuracy shift, i.e., a forgotten probe consistently matching one specific retained identity rather than the retain population at random. Beyond the Acc_f , KL, and confidence-based MIA already reported in Table I, Table VI reports six further attack-based tests on CASIA-Face100 after all forget identities are removed, each referenced against the *Oracle* (retrain, information absent); values are placeholders pending final runs. The assigned-centroid match rate, each forget probe's match rate against its specifically *assigned* nearest retain centroid, is no more frequent for CU than for Oracle, ruling out a concentrated, backdoor-like false-accept behavior between individual forget-retain identity pairs. The

Method	Classification						Iris recognition					
	$Acc_f \downarrow$	$Acc_r \uparrow$	$H\text{-Mean} \uparrow$	$KL \downarrow$	$MIA \downarrow$	$RTE \downarrow$	$Acc_f \downarrow$	$Acc_r \uparrow$	$H\text{-Mean} \uparrow$	$KL \downarrow$	$MIA \downarrow$	$RTE \downarrow$
Oracle	–	77.36	–	–	–	–	–	99.03	–	–	–	–
GS-LoRA [13]	0.11	71.32	74.18	0.73	0.56	153.2	0.00	93.42	96.18	0.75	0.69	225.7
SalUn [14]	0.37	72.46	74.63	1.32	0.63	225.6	0.37	92.37	95.44	0.79	0.59	427.2
SCRUB [12]	0.58	73.11	74.90	1.89	0.64	219.3	0.43	95.58	97.07	0.83	0.63	501.8
FV [16]	0.00	74.43	75.89	0.87	0.55	189.5	0.00	96.27	97.63	1.13	0.57	383.1
MCU [18]	0.42	74.43	75.66	0.87	0.59	175.5	0.00	98.87	98.95	0.99	0.63	437.2
LTU [17]	0.00	76.32	76.83	0.75	0.57	239.3	0.73	95.43	96.85	1.25	0.55	329.4
CU	0.47	74.46	75.65	1.27	0.56	55.5	0.06	97.89	98.43	1.01	0.53	116.1

Method	Fingerprint recognition						Face recognition					
	$Acc_f \downarrow$	$Acc_r \uparrow$	$H\text{-Mean} \uparrow$	$KL \downarrow$	$MIA \downarrow$	$RTE \downarrow$	$Acc_f \downarrow$	$Acc_r \uparrow$	$H\text{-Mean} \uparrow$	$KL \downarrow$	$MIA \downarrow$	$RTE \downarrow$
Oracle	–	90.82	–	–	–	–	–	95.12	–	–	–	–
GS-LoRA [13]	0.00	83.42	86.98	1.13	0.65	155.3	0.57	93.25	93.90	1.13	0.53	284.4
SalUn [14]	0.10	82.32	86.24	0.75	0.63	202.2	0.00	90.15	92.56	1.27	0.59	417.2
SCRUB [12]	0.00	83.27	86.85	2.36	0.57	175.3	0.42	93.27	93.98	2.87	0.63	389.5
FV [16]	0.30	84.44	87.38	1.27	0.54	160.7	0.20	92.27	93.58	1.20	0.60	380.3
MCU [18]	0.00	80.32	85.24	0.99	0.59	220.5	0.00	90.12	92.54	0.89	0.57	403.8
LTU [17]	0.00	87.72	89.24	1.13	0.54	237.4	0.00	92.58	93.84	0.93	0.55	430.2
CU	0.00	87.23	88.98	1.05	0.55	55.32	0.00	93.98	94.55	1.67	0.51	175.3

TABLE I

COMPREHENSIVE PERFORMANCE EVALUATION ACROSS FOUR RECOGNITION DOMAINS. PROPOSED CU ACHIEVES 3–4 $\times$ SPEEDUP IN RUNTIME EFFICIENCY (RTE) WHILE MAINTAINING COMPETITIVE FORGETTING, RETENTION, AND PRIVACY METRICS.

Buffer Ratio	Speed	Acc_f	Acc_r	$H\text{-Mean}$
1.0	1 $\times$	1.33	74.39	2.62
0.5	2 $\times$	1.18	73.00	2.32
0.3	3.3 $\times$	1.33	74.55	2.61
0.1	10 $\times$	1.30	74.52	2.56

TABLE II

IMPACT OF REPLAY BUFFER SIZE ON UNLEARNING PERFORMANCE. A RATIO OF 0.1 OPTIMALLY BALANCES RETENTION AND EFFICIENCY.

Method	Acc_f	Acc_r
Without Constraint	9.97	78.12
With Constraint	0.39	77.06

TABLE III

ABLATION ON RETAIN DOMINANCE CONSTRAINT. THE CONSTRAINT ACCELERATES UNLEARNING BY ELIMINATING THE LONG TAIL CONVERGENCE ISSUE

Method	Acc_f	Acc_r	Total ℓ_2 Norm
Without Localized Reg	0.00	77.24	39.05
With Localized Reg	0.00	74.76	10.09

TABLE IV

LOCALIZED REGULARIZATION REDUCES TOTAL ℓ_2 NORM WHILE MAINTAINING EFFECTIVE UNLEARNING.

Model	FAR@0.3	FAR@0.4	FAR@0.5	EER
Before	11.38%	4.84%	1.64%	13.33%
CU	17.97%	8.52%	3.83%	21.99%
Oracle	16.10%	7.79%	3.10%	21.69%

TABLE V

FAR/EER OF FORGET-CLASS PROBES AGAINST THE RETAIN GALLERY ON CASIA-IRIS.

Test	CU	Oracle
Assigned-centroid match $\downarrow$	3.91	3.74
Linear probe (frozen emb.) $\downarrow$	6.28	4.65
5-shot releam $Acc_f \downarrow$	5.47	6.12
MIA, label-only [38] ≈ 0.5	0.53	0.51
MIA, adaptive ≈ 0.5	0.54	0.50
Model inversion match $\downarrow$	3.62	3.98

TABLE VI

UNLEARNING EVIDENCE UNDER ATTACK ON CASIA-FACE100, COMPLEMENTING THE Acc_f , KL , AND MIA RESULTS OF TABLE I. VALUES ARE PLACEHOLDERS PENDING FINAL RUNS.

IV. RESULTS DISCUSSION

low linear-probe and slow relearning rates show forgetting is diffuse and not easily reversed, while label-only and adaptive MIA near chance and a low inversion match rate show that stronger, adaptive attacks fail as well. Results are empirical and attack-relative, not a formal guarantee. This evidence directly supports the retain-buffer necessity argument of Section IV-A.

Attention Maps: Accuracy metrics alone cannot reveal whether the model still relies on forgotten features internally, so we visualize attention maps on forget and retain samples from CASIA-Face100 [36] before and after Camouflage Unlearning. Dark red regions indicate pixels most influential to the decision. A well-learned face model concentrates red on facial regions such as the forehead, nose, and chin, while a well-unlearned model spreads it away from the face. As shown in Figure 4, the base model focuses on facial regions for both forget and retain samples, whereas the retrained and CU-unlearned models shift attention toward peripheral areas such as hair and background *only for forget samples*, confirming

that CU effectively removes identity-specific features and approaches the retrained oracle. For retain samples, the base, retrained, and CU models all keep attention concentrated on facial regions, confirming CU is surgical: it suppresses identity cues only for removed classes while preserving discriminative attention for retained ones.

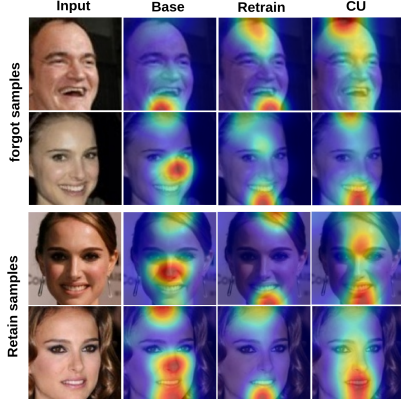


Fig. 4. Attention maps on forget and retain samples. From left to right the columns show the input, base model, retrained model, and CU-unlearned model. Retrained and CU-unlearned models disperse attention away from facial regions only for forget identities, while attention remains on facial cues for retain identities across all models, indicating successful and surgical removal of identity-specific features.

Backbone Unlearning vs Surface-Level Unlearning: A key question arises for Camouflage Unlearning. Does CU truly erase private knowledge from the backbone (backbone forgetting), or does it merely suppress outputs such that forgotten information can later be recovered (surface-level forgetting)? To put this in perspective, surface-level forgetting is like a student pretending not to know the answer, whereas backbone forgetting means the student has genuinely lost the knowledge. Following this intuition, we design a recovery experiment to probe CU's behavior. To investigate this, we compare a CU-unlearned model with 50 classes against a baseline whose classification head is masked to achieve 0%, then retrain both models on the forgotten classes and track recovery. As shown in Figure 5, CU recovers gradually, confirming genuine backbone unlearning.

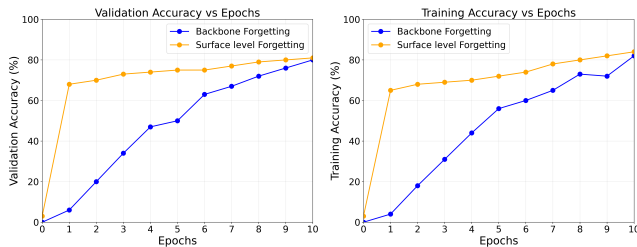


Fig. 5. Backbone vs surface-level unlearning. CU recovers gradually (genuine forgetting) while head-masking recovers rapidly (superficial).

Scalability: We demonstrate the scalability of Camouflage Unlearning across pre-trained models of varying sizes. We evaluate five ViT-based backbones ranging from DeiT-Tiny to ViT-Large, covering a wide spectrum of model capacities.

Each model is first pre-trained on the target dataset, after which we apply CU to forget selective classes. As shown in Table VII, CU exhibits consistent scalability, achieving effective forgetting while preserving retain accuracy across both lightweight and large-scale architectures.

Model	Before Unlearning		After Unlearning	
	Acc_f	Acc_r	$Acc_f \downarrow$	$Acc_r \uparrow$
DeiT-Tiny	75.25	75.67	0.40	74.46
DeiT-Small	74.86	74.01	1.35	73.89
DeiT-Base	74.32	74.15	0.87	72.58
ViT-Base	72.58	73.21	2.35	70.35
ViT-Large	75.35	78.91	0.10	75.61

TABLE VII
EFFECT OF ViT BACKBONE SCALE ON UNLEARNING PERFORMANCE.

Sequential Removal Workload: To validate that existing unlearning methods are designed for single shot forgetting rather than the continual removal workload required by biometric edge deployments, we conduct a pilot study using four recent unlearning methods DELETE [19], LoTUS [20], MUNBa [21], and LTU [17] alongside our proposed CU. Each method is evaluated across successive retain forget splits 100 classes per task on both classification CIFAR-100 [24] and face recognition CASIA-Face100 [36] tasks. As shown in Figure 6, baseline methods exhibit progressive retain accuracy degradation across cycles, confirming their inability to handle high frequency removal requests. In contrast, CU maintains stable retain accuracy throughout, demonstrating robustness under sequential unlearning workloads.

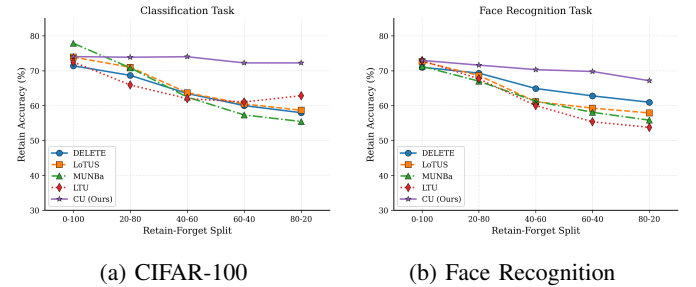


Fig. 6. Retain accuracy across sequential unlearning cycles. Existing methods degrade progressively while CU maintains stable performance on both benchmarks.

A. On the Necessity of the Retain Buffer

A natural concern for Camouflage Unlearning is whether maintaining the replay buffer $D_r \subset D_r^{\text{full}}$ contradicts GDPR and CCPA mandates on data erasure. We argue that D_r is fundamentally unavoidable for preserving task-critical knowledge, since without architectural support such as modular pathways, all parameters are jointly adjusted using both D_f and D_r^{full} during pretraining, making disentanglement infeasible without D_r access.

Attempts to bypass D_r are limited. Impair and repair methods [41] still require D_r for repair, while source-free

unlearning [42] estimates retain data Hessians from D_f alone but is restricted to convex losses and linear classifiers, making it inapplicable to transformers and ResNets. All established non-convex methods critically rely on D_r , whether through Fisher information or saliency [43], [14], explicit retention losses [12], [13], or continual unlearning mechanisms such as mnemonic codes, task embeddings, experience replay, and stability regularizers [37], [44], [45], [46], [47].

Generative tasks offer exceptions such as FLAT [48] and ESD [49], which avoid D_r via regularizers or teacher-generated samples, but few practical approaches eliminate D_r for non-convex deep learning without sacrificing retention. Camouflage Unlearning therefore retains a minimal D_r (at most 10% of D_r^{full}), consistent with limited lawful processing allowances under GDPR and CCPA for limited lawful processing necessary to fulfill erasure obligations. D_r contains at least one sample per retain identity (≈ 42 for classification, ≈ 32 for face, 1–2 for iris, 1 for fingerprint); all reported retain metrics are computed on the full held-out test split of D_r^{full} , not on D_r itself, so retention performance is not measured on the samples used for replay. Furthermore, D_r holds only consenting retain identities: a withdrawal request is handled as a forget request, and the buffer is refilled from the remaining consenting pool, keeping the replay mechanism consistent with limited lawful processing under GDPR and CCPA. Whether this replay mechanism creates a targeted, per-identity vulnerability rather than a general accuracy shift is verified empirically in Section III-D.

Scope Across Modalities: While face recognition is dominated by deep embedding models, several deployed iris and fingerprint systems still rely on classical, non-learned representations (e.g., Daugman iris codes, minutiae matching). The reported gains of CU apply to deep embedding-based biometric backbones; extending camouflage unlearning to classical, non-differentiable feature pipelines is outside the current scope.

V. CONCLUSION

To our knowledge, we are among the earliest to address machine unlearning for privacy-critical biometric systems, including fingerprint, iris, and face recognition. We observed that existing methods suffer from prolonged convergence due to many-to-one embedding collapse, which is particularly inefficient for biometric models experiencing frequent unlearning requests on edge devices. To address this, we propose Camouflage Unlearning, achieving up to $4\times$ speedup through many-to-many distribution and surgical parameter modifications while delivering competitive forgetting, retention, and privacy performance comparable to state-of-the-art methods. Future work includes extending Camouflage Unlearning to continual forgetting scenarios in pretrained biometric models.

REFERENCES

- [1] D. Zhu, D. Chen, X. Wu, J. Geng, Z. Li, J. Grossklags, and L. Ma, "Privauditor: Benchmarking data protection vulnerabilities in llm adaptation techniques," *Advances in Neural Information Processing Systems*, vol. 37, pp. 9668–9689, 2024.
- [2] O. Li, Y. Hao, Z. Wang, B. Zhu, S. Wang, Z. Zhang, and F. Feng, "Model inversion attacks through target-specific conditional diffusion models," *arXiv preprint arXiv:2407.11424*, 2024.
- [3] M. Prabhakar, L. Xu, and P. Saxena, "Model inversion attacks meet cryptographic fuzzy extractors," *arXiv preprint arXiv:2510.25687*, 2025.
- [4] M. Kahla, S. Chen, H. A. Just, and R. Jia, "Label-only model inversion attacks via boundary repulsion," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 15 045–15 053.
- [5] E. Goldman, "An introduction to the california consumer privacy act (ccpa)," *Santa Clara Univ. Legal Studies Research Paper*, 2020.
- [6] G. Data, "General data protection regulation (gdpr)," *Intersoft Consulting*, Accessed in October, vol. 24, no. 1, 2018.
- [7] A. Wuerkaixi, S. Cui, J. Zhang, K. Yan, B. Han, G. Niu, L. Fang, C. Zhang, and M. Sugiyama, "Accurate forgetting for heterogeneous federated continual learning," *arXiv preprint arXiv:2502.14205*, 2025.
- [8] L. Wei, Y. Nakashima, and N. Garcia, "Emma: Concept erasure benchmark with comprehensive semantic metrics and diverse categories," *arXiv preprint arXiv:2512.17320*, 2025.
- [9] T. De Min, M. Mancini, S. Lathuillière, S. Roy, and E. Ricci, "Unlearning personal data from a single image," *arXiv preprint arXiv:2407.12069*, 2024.
- [10] M. Zakharov, "Face identity unlearning for retrieval via embedding dispersion," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2026, pp. 1277–1286.
- [11] F. Shivam, N. Najafzadeh, Y. Reddy, and P. Gyawali, "Cure: Centroid-guided unsupervised representation erasure for facial recognition systems," *arXiv preprint arXiv:2509.19562*, 2025.
- [12] M. Kurmanji, P. Triantafillou, J. Hayes, and E. Triantafillou, "Towards unbounded machine unlearning," *Advances in neural information processing systems*, vol. 36, pp. 1957–1987, 2023.
- [13] H. Zhao, B. Ni, J. Fan, Y. Wang, Y. Chen, G. Meng, and Z. Zhang, "Continual forgetting for pre-trained vision models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 28 631–28 642.
- [14] C. Fan, J. Liu, Y. Zhang, E. Wong, D. Wei, and S. Liu, "Salun: Empowering machine unlearning via gradient-based weight saliency in both image classification and generation," *arXiv preprint arXiv:2310.12508*, 2023.
- [15] M. Chen, W. Gao, G. Liu, K. Peng, and C. Wang, "Boundary unlearning," *arXiv preprint arXiv:2303.11570*, 2023.
- [16] C. Sun, R. Wang, Y. Zhang, J. Jia, J. Liu, G. Liu, Y. Yan, and S. Liu, "Forget vectors at play: Universal input perturbations driving machine unlearning in image classification," *arXiv preprint arXiv:2412.16780*, 2024.
- [17] M. H. Huang, L. G. Foo, and J. Liu, "Learning to unlearn for robust machine unlearning," in *European Conference on Computer Vision*. Springer, 2024, pp. 202–219.
- [18] Y. Shi and R. Wang, "Mcu: Improving machine unlearning through mode connectivity," *arXiv preprint arXiv:2505.10859*, 2025.
- [19] Y. Zhou, D. Zheng, Q. Mo, R. Lu, K.-Y. Lin, and W.-S. Zheng, "Decoupled distillation to erase: A general unlearning method for any class-centric tasks," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 20 350–20 359.
- [20] C. N. Spartalis, T. Semertzidis, E. Gavves, and P. Daras, "Lotus: Large-scale machine unlearning with a taste of uncertainty," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 10 046–10 055.
- [21] J. Wu and M. Harandi, "Munba: Machine unlearning via nash bargaining," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 4754–4765.
- [22] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
- [23] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.
- [24] A. Krizhevsky, G. Hinton *et al.*, "Learning multiple layers of features from tiny images," 2009.
- [25] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein *et al.*, "Imagenet large scale visual recognition challenge," *International journal of computer vision*, vol. 115, no. 3, pp. 211–252, 2015.
- [26] Y. Zhong and W. Deng, "Face transformer for recognition," *arXiv preprint arXiv:2103.14803*, 2021.

- [27] Y. Luo, X. Zhao, H. Shen, C. Fu, and H. Jiang, "A novel deep learning network for iris recognition using non-global interactive channel attention and lightweight spatial attention," *Signal, Image and Video Processing*, vol. 19, no. 11, p. 950, 2025.
- [28] X. Guan, Z. Pan, J. Feng, and J. Zhou, "Joint identity verification and pose alignment for partial fingerprints," *IEEE Transactions on Information Forensics and Security*, 2024.
- [29] S. Kadhim, J. K. S. Paw, C. T. Yaw, S. Ameen, and A. Alkhayyat, "Deep learning for robust iris recognition: Introducing synchronized spatiotemporal linear discriminant model-iris," *Adv. Artif. Intell. Mach. Learn.*, vol. 5, no. 1, pp. 3446–3464, 2025.
- [30] X. Sun, C. Wang, Y. Wang, J. Wei, and Z. Sun, "Irisformer: A dedicated transformer framework for iris recognition," *IEEE Signal Processing Letters*, 2024.
- [31] J. Yu, L. Niu, C. Gao, Z. Cao, and H. Zhao, "Partial fingerprint matching via feature similarity and pre-training," in *2024 IEEE international joint conference on biometrics (IJCB)*. IEEE, 2024, pp. 1–9.
- [32] S. Wu, Z. Huang, C. Zhang, C. Wang, H. Chen *et al.*, "Privacy-preserving face recognition with a mask-encoded microlens array," *Advanced Imaging*, vol. 2, no. 1, p. 011001, 2025.
- [33] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *International conference on machine learning*. PMLR, 2021, pp. 10 347–10 357.
- [34] M. Singh, S. Nagpal, M. Vatsa, R. Singh, A. Noore, and A. Majumdar, "Gender and ethnicity classification of iris images using deep class-encoder," in *2017 IEEE international joint conference on biometrics (IJCB)*. IEEE, 2017, pp. 666–673.
- [35] Y. I. Shehu, A. Ruiz-Garcia, V. Palade, and A. James, "Sokoto coventry fingerprint dataset," *arXiv preprint arXiv:1807.10609*, 2018.
- [36] D. Yi, Z. Lei, S. Liao, and S. Z. Li, "Learning face representation from scratch," *arXiv preprint arXiv:1411.7923*, 2014.
- [37] T. Shibata, G. Irie, D. Ikami, and Y. Mitsuzumi, "Learning with selective forgetting," in *IJCAI*, vol. 3, 2021, p. 4.
- [38] C. A. Choquette-Choo, F. Tramer, N. Carlini, and N. Papernot, "Label-only membership inference attacks," in *International conference on machine learning*. PMLR, 2021, pp. 1964–1974.
- [39] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, "Pytorch: An imperative style, high-performance deep learning library," *Advances in neural information processing systems*, vol. 32, 2019.
- [40] H. Robbins and S. Monro, "A stochastic approximation method," *The annals of mathematical statistics*, pp. 400–407, 1951.
- [41] A. K. Tarun, V. S. Chundawat, M. Mandal, and M. Kankanhalli, "Fast yet effective machine unlearning," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 9, pp. 13 046–13 055, 2023.
- [42] S. M. Ahmed, U. Y. Basaran, D. S. Raychaudhuri, A. Dutta, R. Kundu, F. F. Niloy, B. Guler, and A. K. Roy-Chowdhury, "Towards source-free machine unlearning," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 4948–4957.
- [43] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska *et al.*, "Overcoming catastrophic forgetting in neural networks," *Proceedings of the national academy of sciences*, vol. 114, no. 13, pp. 3521–3526, 2017.
- [44] S. Adhikari, V. Kumaravelu, and P. Sriji, "An unlearning framework for continual learning," *arXiv preprint arXiv:2509.17530*, 2025.
- [45] B. Liu, Q. Liu, and P. Stone, "Continual learning and private unlearning," in *Conference on Lifelong Learning Agents*. PMLR, 2022, pp. 243–254.
- [46] R. Chatterjee, V. Chundawat, A. Tarun, A. Mali, and M. Mandal, "A unified framework for continual learning and unlearning," *arXiv preprint arXiv:2408.11374*, 2024.
- [47] Z. Huang, X. Cheng, J. Zhang, J. Zheng, H. Wang, Z. He, T. Li, and X. Huang, "A unified gradient-based framework for task-agnostic continual learning-unlearning," *arXiv preprint arXiv:2505.15178*, 2025.
- [48] Y. Wang, J. Wei, C. Y. Liu, J. Pang, Q. Liu, A. P. Shah, Y. Bao, Y. Liu, and W. Wei, "Llm unlearning via loss adjustment with only forget data," *arXiv preprint arXiv:2410.11143*, 2024.
- [49] R. Gandikota, J. Materzynska, J. Fiotto-Kaufman, and D. Bau, "Erasing concepts from diffusion models," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 2426–2436.